# Abraham Gonzalez

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## Education

**University of California, Berkeley** August 2018 - August 2025
*Ph.D. in Electrical Engineering and Computer Sciences*, advised by Krste Asanović and Bora Nikolić, GPA: 3.96/4.0
Dissertation Title: "End-to-end Heterogeneous System Design for Hyperscale Big-Data Processing"

**The University of Texas at Austin** August 2014 - May 2018
*B.S. in Electrical and Computer Engineering*, GPA: 3.98/4.0

## Recent Work Experience

**Software Engineer** October 2025 - Present
Google — Sunnyvale, CA

- Software engineer in the AI and Systems Research group led by Hank Levy, researching AI-assisted methodologies for computer architecture discovery and hardware-software co-design for next-generation silicon.
- Collaborated on first-of-their-kind evolutionary AI-assisted microarchitecture discovery methodologies for winning academic cache replacement and prefetching championships.
- Publicly facing research, done in collaboration with the University of California, Berkeley, has been mentioned or featured across blogs (i.e., Google DeepMind AlphaEvolve, Google Antigravity, and ACM SIGARCH blogs), podcasts (i.e., the Computer Architecture Podcast), keynotes (i.e., ASPLOS 2026 keynote by Engineering Fellow and Vice President Partha Ranganathan), and published work.
- Lead organizer of the MICRO 2026 $A^3$: Agentic Approaches to Architecture Workshop and CHIA Hackathon and organizer of the ICLAD 2026 GenAI Chip Hackathon.

**Graduate Student Researcher** August 2018 - August 2025
University of California, Berkeley — Berkeley, CA

- Advised by Professor Emeritus and Professor of the Graduate School Krste Asanović and National Semiconductor Distinguished Professor Borivoje Nikolić.
- Ph.D. candidate and ADEPT/SLICE lab member researching hyperscale cloud architectures, accelerator scheduling, and hardware design methodologies.
- Co-lead of the Hyperscale system-on-chip (SoC) project focused on hyperscale hardware/software co-design.
- Co-lead of the Chipyard SoC framework (used in 20+ tape-outs and 150+ pubs. @ 65+ companies/universities).
- Co-lead of the award-winning FireSim FPGA-accelerated simulator (used in 60+ pubs. @ 25+ companies/univs.).
- Developer of the Berkeley Out-of-Order Machine (BOOM), the first open-source RISC-V out-of-order core.
- Led the tape-out/bring-up of the first Chipyard test chip, a 106.1 GOPS/W heterogeneous SoC in Intel 22FFL.
- Published research and tape-outs at top conferences including ISCA, DAC, IEEE MICRO, and ESSCIRC.
- Lead organizer for 10+ tutorials and workshops with 200+ unique attendees at top conferences.

**Student Researcher Intern** June 2021 - July 2024
Google — Sunnyvale, CA

- Student researcher working with Parthasarathy Ranganathan (Vice President and Engineering Fellow).
- Collaborated with the SystemsResearch@Google (SRG) and Systems Infrastructure Performance teams.
- Researched data processing and remote procedure call (RPC) optimizations as part of the Hyperscale SoC project.
- Published first of its kind research on hyperscale big data processing characterization at ISCA '23.
- Open-sourced HyperRPCBench, a novel representative RPC benchmark suite, with the Fleetbench team.

**Silicon Engineering Group Intern** June 2020 - August 2020
Apple — Cupertino, CA

- Engineering intern working with Si-En Chang (Distinguished Engineer).
- Developed computer architecture tooling for CPU verification.

## Skills

Scala, C/C++, Python, Chisel, SQL, {System}Verilog, RISC-V, Bash, Make, Bazel, TCL, TensorFlow, PyTorch

Git, MapReduce, AWS, Google Cloud, Xilinx Virtex/UltraScale+ FPGAs, Cadence EDA tooling

## Honors, Awards, and Selections

**UC Berkeley:** DARPA Riser (Fa'22), Analog Devices Outstanding Designer (Sp'20), Berkeley Fellowship (Fa'18), EECS Excellence Award (Fa'18), GEM Fellowship (Sp'18), NSF GRFP Honorable Mention (Sp'18)
**UT Austin and prior:** Highest Honors (Sp'18), Distinguished College Scholar (Sp'17/18), College Scholar (Sp'16), Roberto Rocca Scholarship (Fa'17), Victor L. Hand Scholarship (Fa'16), TI Diversity Scholarship (Fa'15), 2nd Best Poster SHPE National Conference (Fa'17), Qualcomm DECA Attendee (1/51 selected nationally) (Sp'15), EOE/SHPE Freshman Academic Excellence Winner (Sp'15)